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APPLIED COMPUTER TECHNOLOGY FOR MANAGEMENT, SYNTHESIS, VISUALIZATION, AND EXPLORATION OF PARAMETERS IN LARGE MULTI-PARAMETER DATA SETS

Abstract

Computer technology is disclosed that applies innovative data processing and visualization techniques to large multi-parameter data sets such as cellular gene expression data to find new relationships such as relationships between cells and genes and create new associative data structures within the data sets that represent these relationships. For example, scatterplots of gene expression data can be iteratively pivoted between a cell view and a gene view to find cell populations and gene sets of interest to a user.

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Background/Summary

CROSS-REFERENCE AND PRIORITY CLAIM TO RELATED PATENT APPLICATION [0001]
This patent application is a continuation of U.S. patent application Ser. No. 15/838,724, filed Dec. 12, 2017, and entitled “Applied Computer Technology for Management, Synthesis, Visualization, and Exploration of Parameters in Large Multi-Parameter Data Sets”, which claims priority to U.S. provisional patent application Ser. No. 62/433,930, filed Dec. 14, 2016, and entitled “Applied Computer Technology for Management, Synthesis, Visualization, and Exploration of Parameters in Large Multi-Parameter Data Sets”, the entire disclosure of each of which is incorporated herein by reference.

INTRODUCTION

[0002] The volume of genetic and gene expression information that is available for both bulk populations and individual cells has grown to the point where it has become unwieldy for investigators. For example, cellular gene expression data can include gene expression data for thousands of genes (e.g., 10,000-30,000 or more genes), can now be measured for individual cells, and thousand cells can be measured per sample. This has presented a tremendous technical problem in the art of visualizing, analyzing, exploring, and making sense of cellular gene expression data.

[0003] For example, with conventional approaches to using computers to facilitate visualization of cellular gene expression data, the visualization is a final end point and the visualization is reached as a result of a user manually writing scripts using the R programming language, which requires the user to have knowledge of different libraries in order to perform data input, reformatting, manipulations, calculations and graphing. These scripts usually have to be customized for particular data sets, and their creation requires expert knowledge of the programming language, existing libraries, and required inputs to produce results. Moreover, such conventional approaches prevent the deep exploration of heterogeneous cell populations.

[0004] As a solution to this technical problem, the inventors disclose the application of computer technology using innovative scatterplot displays across various dimensions of cellular expression data including cell (or cell population) view scatterplots in which cells are visualized as individual data points (e.g., gene vs gene scatterplots of cells) and gene view scatterplots in which genes are visualized as individual data points (e.g., cell population vs cell population scatterplots of genes). Gating can be performed within these scatterplots to create cell populations and gene sets respectively that can serve as biologically-relevant dimensions to be added as new data objects in a workspace for use in augmenting the cellular gene expression data and opening new avenues for meaningful investigation. By way of contrast, performing such analysis in an isolated, siloed manner on the basis of individual genes quickly becomes unwieldy, whereas the ability to pivot between cell view scatterplots and gene view scatterplots allows users to find biologically-relevant grouping of genes that can then be further investigated as synthetic parameters of a cell view scatterplot.

[0005] As noted above, with conventional visualization systems in the art, the visualization serves as an endpoint in the process and cannot serve as a starting point for further creating further

visualization refinements for further investigations. As an example, samples of immune cells from metastatic melanoma patients may contain T cells, and conventional visualization systems in the art would be able only to identify this subset within the immune cells. However, the innovative computer systems described herein allow for deep exploration and analysis of the T cell subset to identify multiple subsets within T cells, for example, T cells which are “exhausted”, track this state to individual genes which can then be targeted to reverse this exhaustion, activate T cells, and thus possibly stimulate an immune response to eradicate the metastases, as explained in greater detail below with reference to example embodiments.

[0006] Accordingly, through the innovative visualization techniques described herein, computer technology can be applied to cellular gene expression data to find new relationships between cells and genes and create new associative data structures within the cellular gene expression data that represents these relationships.

[0007] Through these and other features, example embodiments of the invention provide significant technical advances in the applied bioinformatics arts.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 discloses an example computer system that can be used to support the innovative data processing and visualization techniques described herein.

[0009] FIG. 2A depicts examples of cellular gene expression data sets.

[0010] FIG. 2B depicts a table view of example cellular gene expression data.

[0011] FIG. 3 depicts an example cell view graph window scatterplot.

[0012] FIG. 4 depicts an example process flow for execution to create a cell view scatterplot.

[0013] FIG. 5 shows an example cell view scatterplot user interface that includes parameter selection menus.

[0014] FIG. 6 shows an example of how gating can be performed within a cell view scatterplot user interface to create a cell population.

[0015] FIG. 7 shows an example user interface that permits a user to spread the zeros of a cell view scatterplot.

[0016] FIG. 8 shows an example cell view scatterplot where parameters other than genes are selected as axis parameters.

[0017] FIG. 9 shows an example of how gating can be performed within the cell view scatterplot of FIG. 8 to create a cell population.

[0018] FIG. 10 shows an example user interface for adjusting display settings in a scatterplot presentation.

[0019] FIG. 11 depicts an example gene view graph window scatterplot including cell population selection menus.

[0020] FIG. 12 depicts an example of how cellular gene expression data can be pivoted to create cell population data for a gene view scatterplot.

[0021] FIGS. 13A and B depict examples of how complementary cell populations can be created and defined in a workspace for use in gene view scatterplots.

[0022] FIG. 14 shows an example of how gating can be performed within a gene view scatterplot user interface to create a gene set.

[0023] FIG. 15 shows an example of how cellular gene expression data can be augmented with gene sets as synthetic parameters.

[0024] FIG. 16 shows an example of a cell view scatterplot where gene sets are presented as options in the parameter selection menus.

[0025] FIGS. 17A and B show an example user interface for viewing, editing, and creating new

gene sets from other gene sets in a workspace.

[0026] FIG. **18A-D** show examples of how a user-selected third dimension can be overlaid on a scatterplot.

[0027] FIG. **19** shows an example of how gating can be performed within the 3D scatterplots of FIGS. **18A-D**.

[0028] FIGS. **20A-D** show example reports can be created through the system.

[0029] FIG. **21** shows an example process flow for execution by the system to switch between cell view and gene view modes.

[0030] FIG. **22** shows an example of how the system can be operated to switch between cell view and gene view modes to support investigations and research into cell data.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

[0031] FIG. **1** discloses an example computer system **100** that can be used to support the innovative data processing and visualization techniques described herein. The example computer system **100** comprises a processor **102**, memory **104**, database **106**, and display **108** that can be in communication with each other over an interconnect technology such as bus **110**.

[0032] Processor **102** can take the form of any processor suitable for performing the operations described herein. For example, the CPU of a laptop or workstation would be suitable for use as processor **102**. It should be understood that processor **102** may comprise multiple processors, including distributed processors that communicate with each other over a network to carry out tasks described herein (e.g., cloud computing processing resources). Memory **104** can take the form of any computer memory suitable for cooperating with processor **102** in the execution of the tasks described herein. It should be understood that memory **104** may take the form of multiple memory devices, including memory that is distributed across a network. Similarly, database **106** can take the form of any data repository accessible to the processor **102** (e.g., a file system on a computer, relational database, etc.), and it should be understood that database **106** may take the form of multiple distributed databases (e.g., cloud storage). Display **108** can take the form of a computer monitor or screen that is capable of generating the visualizations described herein.

[0033] The innovative data analysis and visualization techniques described herein can be performed on cellular gene expression data **112**. Cellular gene expression data **112** can be generated by next-generation sequencing (e.g. for the measurement of RNA-Sequencing (RNA Seq) and single cell RNA sequencing (scRNA-Seq) among other sequencing approaches). However, this is only an example, and other techniques for generating cellular gene expression data **112** may be employed. Additional examples include polymerase chain reaction approaches including digital droplet and reverse transcriptase. Still more examples include RNA measurement by flow cytometry, and microarrays, among others, that produce data files which contain the quantification of DNA and/or RNA, or through software programs that process the raw read data (primary and secondary analysis) to generate the gene expression data files. Yet another example is gene expression data derived from stochastic labeling of a sample. Examples of stochastically labeled gene expression data can be found in U.S. Pat. Nos. 9,567,645 and 8,835,358 and in patent application Ser. No. 15/715,028, filed Sep. 25, 2017, and entitled "Measurement of Protein Expression Using Reagents with Barcoded Oligonucleotide Sequences", the entire disclosures of each of which are incorporated by reference.

[0034] Furthermore, the innovative data analysis and visualization techniques described herein can be applied to data generated from single cells by a variety of means. Single cell analysis may include the stochastic labeling of nucleic acids or proteins or any combination of proteins and nucleic acids. As an example, the innovative data analysis and visualization techniques described herein may be used to analyze quantitative features of protein density or gene expression or any combination thereof. The innovative data analysis and visualization techniques described herein can also provide for an improved visualization of quantitative data generated by stochastic labeling of a variety of proteins or nucleic acids in single cells. Quantitative values of biological populations

(nucleic acids, proteins etc.) in individual cells may be compared with other individual cells, or between cell types or even between methods of generating the data. For example gene expression values may be visualized as a function of the method used of generating the data set. The innovative data analysis and visualization techniques described herein can also be used to compare quantitative data generated by a variety of means described here in ways that provide for the visualization of quantitative biological population data independent of the method of generating such data.

[0035] This data **112** can be characterized as a large multi-parameter data set which poses special technical challenges in terms of difficulty in creating meaningful visualizations, particularly when considered with respect to underlying biology so that biologically-relevant information is meaningfully presented to users in a visual manner. For example, the cellular gene expression data may comprise data for large numbers of individual cells and cell populations, with parameters for each cell or cell population that may stretch into 10,000-30,000 or more parameters. Cellular gene expression data **112** can be read out of files in database **106** and loaded into memory **104** as a plurality of data structures **116** to be manipulated by processor **102** during execution of an analysis and visualization program **114**. Program **114** may comprise processor-executable computer code in the form of a plurality of processor-executable instructions that are resident on a non-transitory computer-readable storage medium such as memory **104**.

[0036] FIG. 2A depicts examples of cellular gene expression data sets where each cell (or cell population) is identified by a Cell ID and is associated with a plurality of parameters, each parameter having an ID and a value in relation to a Cell ID. As indicated, gene expression data for cells is highly dimensional and the number of parameters for each cell may reach 10,000-30,000 or more parameters, either per cell or per population of cells. Examples of parameters in the cellular data include counts of gene expression in the subject cell for a large number of genes. Thus, Parameter 1 for Cell 1 may correspond to Gene 1 and its value can be a count of expressions for Gene 1 in Cell 1. Similarly, Parameter 2 for Cell 1 may correspond to Gene 2 and its value can be a count of expressions for Gene 2 in Cell 1. FIG. 2B depicts a table view of example cellular gene expression data **112**. Each row in the table **200** corresponds to a different cell (see the Cell column), and the various columns labeled Gene 1, Gene 2, etc. correspond to different genes and the table cells identify counts of gene expressions for the correspond genes in each subject cell. This table may also include parameters other than genes. For example, the cellular gene expression data **112** may include data values for parameters such as t-distributed stochastic neighbor embedding (tSNE), principal component analysis (PCA), linear discriminant analysis (LDA), etc. in each table cell, where these data values represent an analysis calculation whose value captures differences for individual cells across n parameters. The cellular gene expression data **112** can be stored in any of a number of formats (e.g., as CSV files, database tables (e.g., as relational data in a relational database), sparse data representations, binary formats, and others).

[0037] FIG. 3 depicts an example graph window (GW) that can be produced through execution of program **114** for presentation to a user via display **108**, where the graph window visualizes Gene 1 vs. Gene 2 with respect to a cell population. This visualization can be referred to as a cell view of the cellular gene expression data **112**. As explained below, gating on a selection of individual cells in the cell view creates a cell population. This graph window presents a scatterplot **300** of gene expressions for two user-selected genes **302** and **304** (TMEM216 and MMP2, respectively in this example) with respect to a cell population in a user-selected file **306**. Each file may correspond to a single sample of cell populations, a single cell population (e.g. one population that has been sorted by flow cytometric active cell sorting and analyzed), or possibly multiple samples which have been concatenated (from different patients, or from a patient at different points in time). Each dot **308** in the scatterplot represents a cell in the cell population of the subject file **306**. The scale of the X-axis and Y-axis in this example identify counts for the corresponding gene. Thus, the position of a dot **308** on the horizontal X-axis identifies a count of how many of the gene MMP2 are present in the

cell corresponding to the subject dot **308**, and the position of the dot **308** on the vertical Y-axis identifies a count of how many of the gene TMEM216 are present in the cell corresponding to the subject dot **308**. Thus, dots **308** in the upper right quadrant of the scatterplot **300** correspond to cells where both TMEM216 and MMP2 are highly expressed, while dots in the lower left quadrant of the scatterplot **300** correspond to cells where both TMEM216 and MMP2 are expressed at a low level. Likewise, the upper left quadrant corresponds to cells where TMEM216 is highly expressed but MMP2 is not, while the lower right quadrant corresponds to cells where MMP2 is highly expressed, but TMEM216 is not. The diagonal where $y=x$ is where cells **308** are positioned if those cells equally express both selected genes. Thus, the distance of a cell **308** away from this diagonal in either direction indicates the extent of differential expression of the selected genes in a given cell. It is expected that for many cell populations, there will be large numbers of cells where the expression of the selected genes for those cells is at the zero level. This leads to a large clustering of dots **308** at the zero levels **310** and **312** on the X-axis and Y-axis respectively. Color coding can be used at **310** and **312** to indicate the density of cells with such zero-level gene expressions.

[0038] FIG. 4 depicts an example process flow for execution as part of program **114** that describes how scatterplot **300** can be generated. At step **400**, the processor creates a data structure in a memory workspace (see **116** in FIG. 1). This data structure can be used for holding cell view data.

[0039] For a first axis, the processor selects a gene in the cellular gene expression data **112** based on user input (step **402**). For example, the processor can respond to user input to select a gene column in table **200**. FIG. 5 shows how the user interface can present a user with a list of genes for each axis. To access a selection menu, a user can select a gene selector **302** or **304** for the Y and X axes respectively. If we assume the first axis in this example is the Y-axis, upon selection of **302**, a parameter selection menu **500** is shown which presents a list of parameters available for selection with respect to the Y-axis. This list can be populated with parameters from the cellular gene expression data **112**. As shown in FIG. 5, the user selection **502** for the Y-axis gene is TMEM216. At step **404**, the processor populates the data structure with a list of cells from the cellular gene expression data **112** (e.g., the cells in the cell column of table **200**). Each listed cell is associated with its count value for the selected first axis gene.

[0040] For a second axis, the processor selects another gene in the cellular gene expression data **112** based on user input (step **406**). For example, the processor can respond to user input to select another gene column in table **200**. FIG. 5 shows an example of a user interface where **304** is selected to access an X-axis parameter selection menu **506** (resulting in selection **508** of MMP2). At step **408**, the processor augments the cell list to add to each cell its associated count value for the selected second axis gene. Thus, at this point, the data structure comprises a list of cells associated with count pairs for the selected genes of the first and second axis; for example the list can comprise a set of vectors {Cell ID 1, Gene 1 Count, Gene 2 Count} for each cell in the cellular gene expression data **112**.

[0041] At step **410**, the process parses the list to find the maximum values for each selected gene (the highest counts). These maximum values are then used by the processor to define the appropriate scales for the X-axis and Y-axis in the scatterplot (step **412**). For example, if the maximum value for the X-axis gene is 10, the X-axis scale can be from 0-10. At step **414**, the processor draws the scatterplot based on the cell list and the defined scales using the cell's associated count values as X,Y coordinates in the scatterplot. The result is a scatterplot **300** as shown by FIG. 3.

[0042] Returning to FIG. 3, a user can access gate creation tools **320** in the user interface to create a gate through which child cell populations are created. For example, the user can access a tool **320** to draw a shape in the scatterplot **300** that encompasses a subset of the cells **308**. FIG. 6 depicts an example gate **600** drawn to capture the cells that have non-zero expressions of the two selected genes. The cells **308** falling inside the drawn shape **600** are gated into their own child cell population, and this cell population can be added to the workspace as a distinct object **602**. To

create the cell population, the processor can translate gate **600** into a plurality of boundary conditions for the gated cell population. For example, with respect to a rectangle shape as shown by gate **600**, the boundary conditions can be all cells with (1) X-axis values between 1 and 10, and (2) Y-axis values between 1 and 8. The cell list data structure (e.g., the set of vectors {Cell ID 1, Gene 1 Count, Gene 2 Count}) can be traversed to find all Cell IDs meeting these criteria, and the data for these cell IDs can populate a new child cell population data structure in the workspace. Also, a corresponding scatterplot **300** of gene expressions for the child cell population can then be presented (where the cell population includes one or more cells **308**). This gating can allow a user to focus on a biologically-interesting cluster of cells **308** in the scatterplot **300**.

[0043] Also, the cell view mode of FIG. **3** permits users to visually compare different samples, including child and parent cell populations using navigation tools **322**. For example, through the back and next buttons in tools **322**, a user can navigate to a scatterplot **300** for a next and previous sample in the workspace. Through the down button in tools **322**, a user can navigate to a child cell population in an analysis hierarchy (e.g., to the child cell population created via gate **600** of FIG. **6**). Also, through an up button (not shown), a user can navigate to a parent cell population in the analysis hierarchy.

[0044] Furthermore, when in cell view mode, it is expected that for many cell populations there will be large numbers of cells where the expression of the selected genes for those cells is at the zero level. This leads to a large clustering of dots **308** at the zero levels **310** and **312** on the X-axis and Y-axis respectively of scatterplot **300** shown by FIG. **3**. Color coding can be used at **310** and **312** to indicate the density of cells with such zero-level gene expressions. However, the inventors believe that the display can be enhanced to provide biologically-relevant information to users in certain circumstances by expanding the visualization of the zero levels for the two selected genes (or gene sets as explained below). To accomplish this, the user interface can include a “spread zeros” user control **700** (such as a check box, button, etc.) as shown in FIG. **7**. This control **700** can be provided on a menu for selecting the axis parameters, although if desired by a practitioner, control **700** could be positioned elsewhere such as somewhere on the user interface shown by FIG. **3**. The top half of FIG. **7** shows the scatterplot when the zeros are not spread. The bottom half of FIG. **7** shows the scatterplot when the spread zeros option is selected via input mechanism **700**.

[0045] As shown by the bottom scatterplot in FIG. **7**, box **702** provides an expanded view of the cells that exhibit zero values with respect to the expression of the X-axis gene (MMP2 in this example). Box **704** provides an expanded view of the cells that exhibit zero values with respect to the expression of the Y-axis gene (TMEM216 in this example). The depth of the zero space along the X and Y axes can be defined as a function of how many dots/cells are at the zero levels for each gene, and the dots/cells can be spread across the zero-space defined by boxes **702** and **704** as a function of the density of cells at each zero location, although other distribution techniques could be used. As should be understood, the lower left quadrant of the bottom scatterplot of FIG. **7** (defined by the overlap of boxes **702** and **704**) comprises the cells that have zero expressions for both the X-axis and Y-axis genes. The lower right quadrant of this scatterplot comprises cells that have zero expressions of the Y-axis gene but positive expressions of the X-axis gene, and the upper left quadrant of this scatterplot comprises cells that have zero expressions of the X-axis gene but positive expressions of the Y-axis gene. The upper right quadrant of this scatterplot comprises the cells with positive expressions of both genes (effectively, the scatterplot shown in the top part of FIG. **7**). As can be seen from the bottom scatterplot of FIG. **7**, the cell population of the upper right quadrant is much sparser than the other quadrants. The inventors believe that the ability to visualize the zero levels in this spread manner can provide users with biologically relevant information (such as an ability to assess gene combinations where unexpected distributions occur-e.g., where the upper right quadrant is not as sparse as would be generally expected).

[0046] The cell view scatterplot can also display cell information for selected parameters in the cellular gene expression data **112** other than genes. As indicated above, the cellular gene expression

data **112** may include parameters from dimensionality reduction such as those resulting from tSNE, LDA, PCA, etc. and quality control parameters (above threshold, parameters relating to ribosomal RNA (rRNA) abundance, etc.) These parameters can be presented as options on the parameter selection menus **500** and **506**. FIG. **8** shows an example where selection **800** for the Y-axis is the parameter tSNE axis 2 and where selection **02** for the X-axis is the parameter tSNE axis 1. The resultant scatterplot of FIG. **8** can be created via the FIG. **4** process flow. A user can also gate desired cell populations in the scatterplot of FIG. **8** (see gate **900** in FIG. **9**) as explained above with respect to FIG. **6**.

[0047] The cell view user interface of FIG. **3** can also include a user control **330** (e.g., the “T” button of FIG. **3**) through which the user can alter the display of information in scatterplot **300**. Upon selection of T button **330**, the user interface of FIG. **10** can be presented. Through this interface, a user can change the binning/display of scatterplot data interactively and live. The list at right in FIG. **10** shows the available and selected parameters for the X-axis of the scatterplot. A histogram **1000** shows the binning of values for the selected parameter with respect to the subject cell population. Given the high prevalence of zeros, histogram **1000** shows a large spike at the zero level, and the scale of this example obscures the non-zero levels in the histogram. Through left/right arrows shown toward the top of FIG. **10**, a user can quickly preview displays for different samples. Through +/- controls in the middle of FIG. **10**, a user can easily alter the histogram zoom level.

[0048] Through scale controls **1002**, the user can adjust the X-axis of the scatterplot in a variety of ways. For example, the X-axis scale can be defined to exhibit a liner scale or some other scale (such as a log2 scale) (see control **1004**). Also, through min/max controls **1006**, the user can define the minimum and maximum boundaries on the X-axis. For example, through these controls, the minimum value can be defined to be a value greater than zero, which would remove the zero spike from the histogram and re-present a newly scaled histogram where the distribution of non-zero values across the X-axis can be more clearly seen. Sliders **1008** can provide users with easy control over the transformation variables. Also, it should be understood that additional transformation options can be provided, including user-supplied transforms whose variables can be adjusted by user input (see, e.g., US Pat App Pub 2016/0328249 entitled “Plugin Interface and Framework for Integrating External Algorithms with Sample Data Analysis Software”, the entire disclosure of which is incorporated herein by reference).

[0049] A particularly innovative and powerful aspect of the inventive system disclosed herein is the ability to pivot the scatterplot display from a cell view mode to a gene view mode. FIG. **11** shows an example of a scatterplot **1100** in the gene view mode. As used herein, a “gene view” visualization refers to a visualization where genes are the individual data points measured against the axis dimensions (e.g., genes as the dots in a scatterplot measured against two cell populations). With scatterplot **1100**, the axis parameters are cell populations (see the X-axis parameter **1102** which identifies a population of B cells, and the Y-axis parameter **1104** which identifies a population of “not B” cells in this example). The dots **1106** in scatterplot **1106** represent specific genes (rather than individual cells as in scatterplot **300** of FIG. **3**). Population selection menu **1110** provides a list of cell populations available for selection to be used on the X-axis, and population selection menu **1112** provides a list of cell populations available for selection to be used on the Y-axis. As noted, in this example, selections **1114** and **1116** correspond to populations of B cells and “not B” cells respectively.

[0050] With this pivot, with reference to cellular gene expression data **112** such as table **200** from FIG. **2**, the table **200** can be pivoted such the genes become rows in the table and subsets of the cells are grouped into two cell populations that become the columns of the tables. Furthermore, computations can be performed on the gene counts in table 2 for the cells of the two cell populations to determine the values that will populate the cells of the pivoted table. Through control **1120** of FIG. **11**, a user can define the computations that are to be performed on the gene

data to compute the values for the cells in the pivoted table. In the example of FIG. 11, a normalized mean computation has been selected. However, it should be understood that other computational options can be available, such as normalized medians, normalized modes, straight means, straight medians, straight modes, etc. Accordingly, it should be understood that the computation performed on the pivoted table data can be any user-defined function that is deemed biologically-relevant to a user. The inventors note that it may be desirable for the computation to be based on averaging of some sort to account for potential discrepancies in the cell counts of the two cell populations and/or normalization to account for the variability between samples (e.g. normalization to a spike-in of External RNA Controls Consortium (ERCC) controls which contain a known amount of RNA to be measured). For example, if the pivoted table values were straight counts of each gene in the two cell populations, and if there was a meaningful difference in the count of cells in the two cell populations, the aggregated gene count totals across the two cell populations would not be very informative in a comparative sense. However, if the cell populations were roughly similar in size, straight gene counts for each gene in the two cell populations might nevertheless be informative.

[0051] FIG. 12 depicts an example pivot from a cell view to a gene view. Table 200 in FIG. 12 shows gene expressions per cell for a number of cells in a sample. Each row corresponds to a different cell, and the columns correspond to expression counts for different genes in the associated cells. If subsets of these cells are grouped into two cell populations 1210 and 1212 as shown in FIG. 12 (e.g., via gating in the cell view scatterplot 300), these two cell populations can be pivoted as shown to create the pivoted table 1200. In pivoted table 1200, the rows are the genes that were columns in table 200. The columns in table 1200 are the two cell populations 1210 and 1212. Each table cell is populated with a concatenation of the gene counts for the associated gene in the cells of each cell population. In the example of FIG. 12, the computation being performed on these values is a straight averaging to compute means as shown in pivoted table 1200a. Once again, as noted above, it should be understood that other computations could be performed if desired by a practitioner. A gene view scatterplot of the type shown by scatterplot 1100 in FIG. 11 can then be created from pivoted table 1200a using the same basic techniques described in FIG. 4 (albeit using pivoted table 1200a rather than table 200).

[0052] Returning to FIG. 11, the gene view scatterplot 1100 thus provides users with a powerful manner of visualizing differential gene expression between two cell populations over a list of genes that may be extremely large in size. The inventors believe that scatterplot 1100 represents a pioneering new way of visualizing cellular gene expression data in a manner that opens up a wide new array of investigatory options for practitioners, some examples of which are described below. The example gene view scatterplot 1100 of FIG. 11 provides users with flexibility in specifying the x/y axis populations and dynamically updating gene expression graphs as populations change. The example gene view scatterplot 1100 also provides users with a plethora of graph options and resolutions. Importantly, the analysis does not stop with this graph; it continues with the ability to create new gene sets and dig deeper into the data as discussed below. Examining cellular heterogeneity and differences between samples/patients revealed by particularly new single cell methods is not possible without this approach.

[0053] The diagonal where $y=x$ is where genes 1106 are positioned if those genes are equally expressed in both cell populations according to the metric defined via 1120. Thus, the distance of a gene 1106 away from this diagonal in either direction indicates the extent of differential expression of the subject gene 1106 as between two selected cell populations. Given that it is expected that most genes 1106 will not be expressed (or will be only lightly expressed) in many cell populations, it is expected that there will typically be large cluster of genes 1106 in the lower left quadrant of scatterplot 1100.

[0054] Furthermore, since users can create multiple, hierarchically-related cell populations that exist as data objects in the workspace as described in connection with FIG. 6, the user can select

any of these cell populations while in the gene view mode. Furthermore, it should be understood that the workspace can be used to create Boolean expressions of cell populations (e.g., the OR combination of all cell populations that are “Not B Cells” to create complementary cell populations) for exploration in the gene view mode. FIGS. **13A** and **B** show examples of such complementary Boolean cell populations in a workspace. Accordingly, a user can create a scatterplot **1100** that shows differential gene expression as between a population of B cells and a population of “not B” cells.

[0055] Another powerful and innovative aspect of the gene view mode described herein is an ability for users to gate genes while in the gene view mode to thereby create gene sets. FIG. **14** shows an example where gate **1400** is created in the gene view scatterplot to capture a set of genes that the user deems worthy of further investigation. Gate **1400** can be drawn on the scatterplot using the techniques described above in connection with FIG. **6**. This gating will create a data structure in the workspace corresponding to the gene set defined by gate **1400** (where the gene set comprises one or more genes depending on how many genes **1106** are encompassed by the user-defined gate **1400**). Such a gene set can then serve as a synthetic parameter in relation to the cellular gene expression data **112** that can be selected for visualization while in the cell view mode. In the example of FIG. **14**, gate **1400** is a trapezoidal shape that captures the genes **1106** that have a positive differential expression in B cells versus non-B cells. The extent of the positive differential expression is controlled by the user through the distance of the trapezoid's diagonal line away from the $y=x$ diagonal of the scatterplot.

[0056] FIG. **15** shows an example of how the cellular gene expression data **112** can be augmented with gene sets as a synthetic parameter (in the form of example augmented table **200**). This table **200** has been augmented with two gene sets as parameters (see the Gene Set 1 and Gene Set 2 columns). In this example, Gene Set 1 consists of {Gene 2, Gene 3} and Gene Set 2 consists of {Gene 1, Gene 4}. The table cells for these gene sets (each corresponding to a different cell in the table rows) can then be populated with the sums of the gene counts for the constituent genes of the subject gene sets. Accordingly, a user can return to the cell view mode to create a cell view scatterplot where one or both of the X-axis and Y-axis parameters are gene sets. An example of this is shown by FIG. **16**. The parameter selection menus for each axis in FIG. **16** include a section that lists available gene set options. The cell view scatterplot of FIG. **16** thus shows a scatterplot of cells **308** where the X-axis parameter is the gene set labeled “B_GeneTable” and the Y-axis parameter is the gene set labeled “HighTHighB”. Thus, it should be understood that the gating capabilities within the gene view scatterplot to create gene sets in combination with the ability use the created gene sets as a synthetic parameter used in a cell view scatterplot provide users with unprecedented capabilities for intelligently reducing the multi-parameter data space of cellular gene expression data. This combination allows the user to overcome the extremely difficult problem of identifying common genes in populations and identifying the differential genes in populations. In other words, whether the user is looking for what genes are in common, or which genes are different, the ability to create new gene sets and viewing/analyzing them as a single parameter allows the user to focus on the important relationships between cell populations and gene sets. Furthermore, as users create gene sets of biological interest based on various comparisons of cell populations, these gene sets can be shared with other users by passing these gene sets (as a defined list of encompassed genes) to other users so that those other uses can evaluate the gene sets with their samples of cell data. Based on such sharing and independent investigation across different cellular data sets, it is expected that greater insights into gene behavior with respect to cells can be gained.

[0057] FIG. **17A** shows an example view into a workspace that breaks the workspace down into various gene sets **1700**, samples **1702**, and cell populations **1704** that exist within the workspace. As shown, the section **1700** that lists gene sets can include various display fields that provide metadata about each subject gene set (e.g., a name, count of genes within the gene set, and a description of the gene set). The name and description can be editable by a user and will be helpful

for informing users about pertinent characteristics of the gene set. Also, the gene sets can be listed in a manner that indicates any hierarchical relationships that exist within gene sets.

[0058] Gene set controls **1710** (shown in greater detail in FIG. **17B**) provide a user with an ability to create gene sets from other gene sets via Boolean operations. For example, via the union button, a user can create a new gene set that is the union of two or more selected gene sets within list **1700**. Via the intersection button, a user can create a new gene set that consists of the genes that are present in both/all of two or more selected gene sets within list **1700**. Via the complement button, a user can create a complementary gene set from two gene sets (Gene Set 1 and Gene Set 2) within list **1700** where a first complementary new gene set consists of all of the genes that are in Gene Set 1 but not Gene Set 2 and where a second complementary new gene set consists of all of the genes that are in Gene Set 2 but not Gene Set 1. Via the “All Comparisons” button, a user can create several new gene sets at the same time via each of the techniques described above (new gene sets via the union operation, the intersection operation, and the complementary operation). This flexible ability to create new gene sets from existing gene sets allows for the comparison of gene sets between treatment conditions, patients, experiments, etc. Accordingly, by using controls **1710**, a user can create gene collections that comprise combinations of gene sets.

[0059] Another powerful and innovative aspect of the visualizations provided by program **116** include an ability to overlap a third dimension on the cell view and/or gene view scatterplots. For example, color coding (e.g., a heatmap) can be applied to the cells **308** or genes **1106** in a scatterplot to provide another dimension to the data presentation. An example of this is shown by FIGS. **18A-D**. FIG. **18A** shows how third dimension controls **1800** can be used to define how the third dimension is presented within an example cell view scatterplot display (e.g., as a heatmap statistic). Through control **1804**, the user can define the statistic to be used for the heatmap. FIG. **18B** shows the options that are available for this statistic **1804** in an example embodiment (e.g., median, mean, geometric mean, coefficient of variation (CV), robust CV, standard deviation (SD), robust SD, etc.). A heatmap statistic can be calculated as follows: Each dot in the graph represents one or more cells (usually multiple cells). For each dot, the selected statistic (e.g., mean, median, etc) is calculated for the cell(s). Thus if there are 400 dots in the graph, then 400 statistical values are calculated. For the 400 statistical values, the min and max are determined, and are used as the lower and upper bound to index into a color map (an array of color values whose gradient changes from one color to another) i.e the min value is mapped to index 0 in the color array, and the max value is mapped to the last index in the color map. Colors in the color map can then be applied to values between the min and max values in some fashion (such as a linear distribution of colors to values).

[0060] FIG. **18C** shows how a user can select a parameter **1808** from the cellular gene expression data for use as the third dimension. Parameter list **1810** can be populated with a list of the parameters for the cellular gene expression data (e.g., the columns in table **200**). In this example, the user has selected the parameter HighT HighB as a third dimension to be overlaid via color coding on the cell view scatterplot, as shown by FIG. **18D**. This third dimension overlay provides a user with a further capability for meaningfully interpreting a scatterplot. For example, the user can use the color coding to identify cell populations of interest and gate such cell populations (see gate **1900** in FIG. **19**), thereby creating another new object in the workspace as a cell population for further investigation.

[0061] Furthermore, according to another aspect of the disclosed system, reports can be created in a report editor by dragging cell populations from the workspace into the report editor (see FIG. **20A**). As another example, cell populations can be overlaid by dragging one cell population on top of another to create 2D overlaps and heatmaps by gene set. Overlays of cell populations are shown in FIG. **20A-D** in two ways. In the scatter plot, the different populations are shown as dot plots of different colors in the same graph (and you can choose which population is laid on top of the other). In the heat map, the different populations are organized and appended horizontally as

additional columns in the heat map, i.e. all cells of population 1 are rendered together and labeled, followed by all cells of population 2, etc. FIG. 20B shows example heatmaps by gene sets and cell populations. FIG. 20C shows an enlarged view of the leftmost portion of FIG. 20B, and FIG. 20D shows an enlarged view of the rightmost portion of FIG. 20C.

Example Use Case

[0062] As indicated above, the disclosed system provides a powerful mechanism for investigating cellular gene expression data **112** by switching between the cell view mode and gene view mode (or vice versa) while performing gating in those two viewing modes to focus on data of interest.

[0063] FIG. 21 depicts an example process flow for execution as part of program **114** that describes how a user can switch between the cell view mode and gene view mode to support deep investigations of cellular gene expression data. At step **2100**, the processor selects a sample from the workspace in response to user input. At step **2102**, a decision is made regarding whether to operate in cell view mode or gene view mode. This decision can be made in response to user input. If a cell view mode is selected, the process flow proceeds to step **2104**. If a gene view mode is selected, the process flow proceeds to step **2122**. Steps **2104-2120** correspond to operation in the cell view mode, while steps **2122-2140** correspond to operation in the gene view mode.

[0064] At step **2104**, the processor selects cell view axis parameters in response to user input (e.g., selection of parameters such as genes or gene sets). At step **2106**, the processor generates a cell view scatterplot data structure from the cellular gene expression data **112** based on the axis parameters selected at step **2104**. This step may involve selecting the columns corresponding to the selected parameters in table **200** to obtain a list of cells and their associated values for each selected parameter. At step **2108**, the processor generates the cell view scatterplot **300** from the data within the cell view scatterplot data structure created at step **2106** for presentation to the user.

[0065] At step **2110**, the processor receives a gate specification with respect to the cell view scatterplot in response to input from a user. This gating creates a cell population (step **2112**), where the created cell population gets saved as a new data structure in the workspace. At this point, the user can choose whether to (1) work with a new sample (see step **2114** with progression back to step **2100**), (2) define one or more new axis parameters with respect to the current sample while in the cell view mode (see step **2116** with progression back to step **2104**), (3) define a new gate with respect to the current sample while in the cell view mode (see step **2118** with progression back to step **2110**), or (4) switch to the gene view mode (see step **2120** with progression to step **2122**).

[0066] At step **2122**, the processor pivots the cellular genetic expression data **112** as discussed above. Then, at step **2124**, the processor selects cell populations from the workspace in response to user input. At step **2126**, the processor generates a gene view scatterplot data structure from the pivoted cellular gene expression data based on the cell populations selected at step **2124**. This step may involve selecting the pivoted columns corresponding to the selected cell populations in a pivoted version of table **200** to obtain a list of genes and their associated metrics for each selected cell population. At step **2128**, the processor generates the gene view scatterplot **1100** from the data within the gene view scatterplot data structure created at step **2126** for presentation to the user.

[0067] At step **2130**, the processor receives a gate specification with respect to the gene view scatterplot in response to input from a user. This gating creates a gene set (step **2132**), where the created gene set gets saved as a new synthetic parameter in the workspace for association with the cellular genetic expression data. The cellular genetic expression data can thus be augmented with new data values corresponding to the gene set created at step **2132**. At this point, the user can choose whether to (1) work with a new sample (see step **2134** with progression back to step **2100**), (2) define one or more new cell populations with respect to the current sample while in the gene view mode (see step **2136** with progression back to step **2124**), (3) define a new gate with respect to the current sample while in the gene view mode (see step **2138** with progression back to step **2130**), or (4) switch to the cell view mode (see step **2140** with progression to step **2104**).

[0068] Thus, FIG. 21 shows how the system can be used by a user to quickly transition between

cell view modes and gene view modes while creating cell populations and gene sets respectively via gating that can be used to aid visualizations after switching between modes.

[0069] As an example, a powerful and innovative mode of operation is shown by the example process flow of FIG. 22 where a user interacts with program 114 via the process flow of FIGS. 21 to (1) create one or more cell populations within cell view modes of the display (step 2200), (2) pivot to a gene view mode that comparatively displays gene expressions across multiple cell populations (step 2202), (3) create one or more gene sets within gene view mode of the display (step 2204) which augments the cellular gene expression data with the new gene set(s) as a synthetic parameter (step 2206), (4) pivots back to a cell view scatterplot using one or more of these gene sets as axis parameter(s) for a cell view scatterplot (step 2208), and (5) iteratively repeats these operations as desired to drill down into biologically relevant relationships that might exist within the cellular gene expression data 112 (and where operations (1) and (3) may be aided with overlays of user-defined third dimensions on the cell view or gene view data display). For example, the disclosed system can serve as a powerful tool for research into precision/personalized medicine to perform work such as evaluating cell data from numerous patients to find patients with genes that correlate well with survival and therapy response to various cancers or other illnesses/pathologies.

[0070] For example, through the tools provided herein, a user might be able to analyze cell populations to identify differentially expressed gene sets that are correlated to better chances for survival with respect to a particular Cancer X (which we can label as “Survival Gene Set”). At the same time, a user might be able to analyze cell populations to identify differentially expressed gene sets that are correlated to poor chances for survival with respect to Cancer X (which we can label as “Not Survival Gene Set”). Further still, a user might be able to analyze cell populations to identify differentially expressed gene sets that are correlated to responding well to Therapy Y for Cancer X (which we can label “Therapy Responsive Gene Set”). Then, these gene sets can be used as synthetic parameters in the cell view mode of the system to find cell populations in patients that are genetically predisposed to respond well to treatment by Therapy Y in order to survive Cancer X. For example, the cell view scatterplot can use the

[0071] Survival Gene Set as one axis parameter (e.g., X-axis parameter) and the Not Survival Gene Set as the other axis parameter (e.g., Y-axis parameter), while using the Therapy Responsive Gene Set as the third dimensional overlay. The resultant scatterplot can show cell populations that will correlate well with both survival and therapy responsiveness (as well as cell populations that do not correlate well with survival or therapy responsiveness).

[0072] While the invention has been described above in relation to its example embodiments, various modifications may be made thereto that still fall within the invention's scope. For example, while the example scatterplots shown herein present the X-axis as a horizontal axis and the Y-axis as a vertical axis, it should be understood that some practitioners may find it desirable to tilt the scatterplots. An example would a scenario where the diagonal $y=x$ is deemed biologically important. In such a case, the scatterplot might be tilted so that the $y=x$ diagonal is presented as a horizontal or vertical line rather than a 45 degree line to thereby help focus users on how far data might lie away from the $y=x$ line. Accordingly, it should be understood that these and other modifications to the invention will be recognizable upon review of the teachings herein.

Claims

1-24. (canceled)

25. A computer-implemented method of visualizing a multi-parameter data set performed by one or more processors executing program instructions, the method comprising: receiving a multi-parameter data set comprising cellular gene expression data of a plurality of cells generated by flow cytometry equipment, the multi-parameter data set comprising a plurality of data items, wherein

each data item is associated with an expression of a gene; receiving a first input comprising a selection of a first parameter and a second parameter; receiving a second input comprising a selection of a data item from the multi-parameter data set of a first cell population associated with the first parameter and a second cell population associated with the second parameter; receiving a gate creation input defining a gated gene set comprising a subset of a plurality of different genes expressed by the first cell population and the second cell population; calculating a synthetic parameter for each cell of the plurality of cells based on two or more genes from the gated gene set; and displaying the multi-parameter data set with at least one axis corresponding to the synthetic parameter.

26. The method of claim 25, wherein each of the data items comprises data indicative of counts for expressions of the corresponding gene in the cell corresponding to that particular data item.

27. The method of claim 25, further comprising displaying a cell view scatterplot of the multi-parameter data set across first and second axes corresponding to the user-selected first and second parameters, the cell view scatterplot comprising a plurality of dots, wherein each dot in the plurality of dots corresponds to a data item from the multi-parameter data set.

28. The method of claim 27, further comprising: receiving, via a user interface, a user specification of a new first parameter associated with the at least one axis; and updating, within the user interface, the cell view scatterplot based on the user specification of the new first parameter associated with the at least one axis of the cell view scatterplot.

29. The method of claim 25, further comprising, in response to receiving the second input, pivoting from a cell view scatterplot generated in response to the first input, to a gene view scatterplot by displaying the gene view scatterplot comprising a plurality of dots, wherein each dot of the plurality of dots corresponds to one of a plurality of different genes expressed by the first and second cell populations.

30. The method of claim 29, wherein the gate creation input comprises a two-dimensional shape defined within the gene view scatterplot via a graphical user interface.

31. The method of claim 25, further comprising augmenting the cellular gene expression data such that the synthetic parameters are associated with cells in the cellular gene expression data.

32. The method of claim 25, wherein the cellular gene expression data comprises a plurality of stochastic labels derived from stochastic labeling of a sample.

33. An apparatus comprising: a memory configured to store a multi-parameter data set; a processor in communication with the memory, the processor configured to perform the method of claim 25; and a display in cooperation with the processor, wherein the display is configured to graphically present the multi-parameter data set.

34. A computer program product comprising: a plurality of processor-executable instructions that are resident on a non-transitory computer readable storage medium, wherein the instructions are configured, upon execution by a processor, to cause the processor to: receive a multi-parameter data set comprising cellular gene expression data of a plurality of cells generated by flow cytometry equipment, the multi-parameter data set comprising a plurality of data items, wherein each data item is associated with an expression of a gene; receive a first input comprising a selection of a first parameter and a second parameter; receive a second input comprising a selection of a data item from the multi-parameter data set of a first cell population associated with the first parameter and a second cell population associated with the second parameter; receive a gate creation input defining a gated gene set comprising a subset of a plurality of different genes expressed by the first cell population and the second cell population; calculate a synthetic parameter for each cell of the plurality of cells based on two or more genes from the gated gene set; and display the multi-parameter data set with at least one axis corresponding to the synthetic parameter.

35. The computer program product of claim 34, wherein the instructions are further configured to cause the processor to display a cell view scatterplot of the multi-parameter data set across first and

second axes corresponding to the user-selected first and second parameters, the cell view scatterplot comprising a plurality of dots, wherein each dot in the plurality of dots corresponds to a data item from the multi-parameter data set.

36. The computer program product of claim 35, wherein the instructions are further configured to cause the processor to: receive, via a user interface, a user specification of a new first parameter associated with the at least one axis; and update, within the user interface, the cell view scatterplot based on the user specification of the new first parameter associated with the at least one axis of the cell view scatterplot.

37. A system comprising: one or more processors; and a non-transitory computer readable storage medium having stored thereon computer-executable instructions that, upon execution by the one or more processors, cause the one or more processors to: receive a multi-parameter data set comprising cellular gene expression data of a plurality of cells generated by flow cytometry equipment, the multi-parameter data set comprising a plurality of data items, wherein each data item is associated with an expression of a gene; receive a first input comprising a selection of a first parameter and a second parameter; receive a second input comprising a selection of a data item from the multi-parameter data set of a first cell population associated with the first parameter and a second cell population associated with the second parameter; receive a gate creation input defining a gated gene set comprising a subset of a plurality of different genes expressed by the first cell population and the second cell population; calculate a synthetic parameter for each cell of the plurality of cells based on two or more genes from the gated gene set; and display the multi-parameter data set with at least one axis corresponding to the synthetic parameter.

38. The system of claim 37, wherein each of the data items comprises data indicative of counts for expressions of the corresponding gene in the cell corresponding to that particular data item.

39. The system of claim 37, wherein the instructions further cause the one or more processors to display a cell view scatterplot of the multi-parameter data set across first and second axes corresponding to the user-selected first and second parameters, the cell view scatterplot comprising a plurality of dots, wherein each dot in the plurality of dots corresponds to a data item from the multi-parameter data set.

40. The system of claim 39, wherein the instructions further cause the one or more processors to: receive, via a user interface, a user specification of a new first parameter associated with the at least one axis; and update, within the user interface, the cell view scatterplot based on the user specification of the new first parameter associated with the at least one axis of the cell view scatterplot.

41. The system of claim 37, wherein the instructions further cause the one or more processors to, in response to receiving the second input pivot from a cell view scatterplot generated in response to the first input, to a gene view scatterplot by displaying the gene view scatterplot comprising a plurality of dots, wherein each dot of the plurality of dots corresponds to one of a plurality of different genes expressed by the first and second cell populations.

42. The system of claim 41, wherein the gate creation input comprises a two-dimensional shape defined within the gene view scatterplot via a graphical user interface.

43. The system of claim 37, wherein the instructions further cause the one or more processors to augment the cellular gene expression data such that the synthetic parameters are associated with cells in the cellular gene expression data.

44. The system of claim 37, wherein the cellular gene expression data comprises a plurality of stochastic labels derived from stochastic labeling of a sample.
